Project Design Phase-II
Technology Stack (Architecture & Stack)

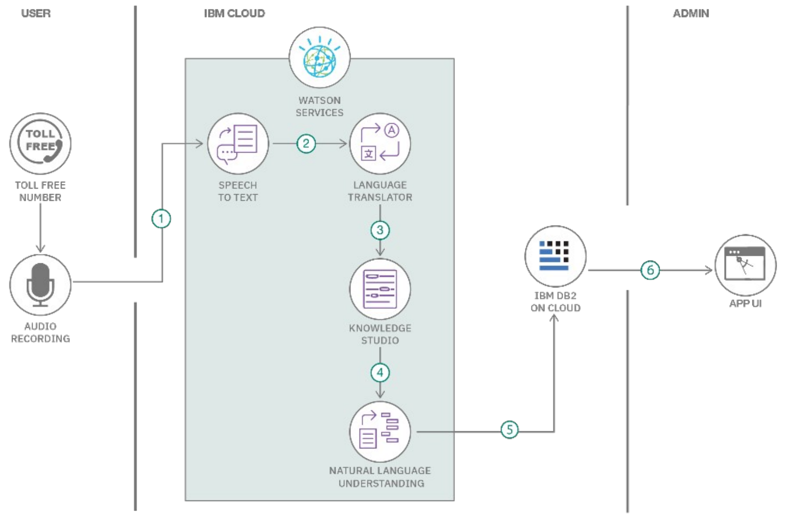
Date	22 July 2026
Team ID	TEAM-HT-2026
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, Bootstrap, Flask (Jinja templates) – hosts the embedded Tableau dashboard and story
2	Application Logic-1	Logic for a process in the application	Python (Flask) – serves the web pages and routes that embed the Tableau views
3	Application Logic-2	Logic for a process in the application	Python (Pandas) – data cleaning, transformation and pre-processing of the UNESCO dataset
4	Application Logic-3	Logic for a process in the application	Tableau calculated fields – Danger Flag, Years Since Inscription, Region Rank
5	Database	Data Type, Configurations etc.	CSV / Excel source file (UNESCO World Heritage Sites 2019 dataset)
6	Cloud Database	Database Service on Cloud	Not used – dataset is a static file loaded directly into Tableau
7	File Storage	File storage requirements	Local filesystem for raw/cleaned CSV and the Tableau workbook (.twbx)
8	External API-1	Purpose of External API used in the application	Not applicable for this project
9	External API-2	Purpose of External API used in the application	Not applicable for this project
10	Machine Learning Model	Purpose of Machine Learning Model	Not applicable – project is descriptive/visual analytics, not predictive

			modeling
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Tableau Public/Server for hosting the workbook; Flask app deployed locally or on a cloud host (e.g. Render/Heroku) for embedding

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used	Flask, Bootstrap, Pandas, Tableau Public (free tier)
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	Tableau Server/Public view-level permissions; Flask input validation on any query parameters used to select dashboard views
3	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Simple 2-tier setup (Flask front-end + Tableau-hosted visualization layer); can extend to additional dashboards/years without redesign
4	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Dashboard availability depends on Tableau Public/Server uptime; Flask app can be deployed on any always-on host
5	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN’s) etc.	Use of Tableau data extracts (.hyper) instead of live connections, and filters to limit the amount of data rendered per view, keeps interaction responsive

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Our Technical Architecture – Heritage Treasures Project

The diagram below shows the end-to-end flow: the raw UNESCO dataset is cleaned in Python, visualized and assembled into a dashboard/story in Tableau Desktop, published to Tableau Server/Public, and finally embedded into a Flask web application for stakeholders to access.

Heritage Treasures — Technical Architecture Flow